A 647911

⇒Final Exam⇐

QUESTION#1

Given:-

Spider-man = x
Hulk = y
Wonder-woman = z

$$x + y + z = 8$$
$$4x + 10y + 12z = 60$$

In augmented matrix form

$$\left[\begin{array}{ccc:c} x & y & z & 8 \\ 4x & 10y & 12z & 60 \end{array}\right] \Rightarrow \left[\begin{array}{ccc:c} 1 & 1 & 1 & 8 \\ 4 & 10 & 12 & 60 \end{array}\right]$$

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & 8 \\ 4 & 10 & 12 & 60 \end{array}\right]$$

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & 8 \\ 0 & 6 & 8 & 28 \end{array}\right] \quad R_2 - 4R_1 \qquad \begin{array}{cccc} 4 & 10 & 12 & 60 \\ -4 & -4 & -4 & -32 \end{array}$$

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & 8 \\ 0 & 1 & \frac{8}{6} & \frac{28}{6} \end{array}\right] \quad \cancel{R_2 - 5R_1} \quad R_2 \times +\frac{1}{6}$$

Here Gauss's elimination is complete

$$\left[\begin{array}{cccc} 1 & 0 & -\frac{1}{3} & : \frac{10}{3} \\ 0 & 1 & \frac{8}{6} & : \frac{28}{6} \end{array}\right] R_1 - R_2$$

$$x - \frac{1}{3}z = \frac{10}{3} \Rightarrow x = \frac{10}{3} + \frac{1t}{3}$$

$$y + \frac{8}{6}z = \frac{28}{6} \Rightarrow y = \frac{28}{6} - \frac{8t}{6}$$

Let $z = t$

Let $t = 2$

for $x = \frac{10}{3} + \frac{1t}{3} = \frac{10+1t}{3} = \frac{10+1(2)}{3}$

$$x = \frac{10+2}{3} = \frac{12}{3} = 4$$

for $y = \frac{28}{6} - \frac{8t}{6} = \frac{28-8t}{6} = \frac{28-8(2)}{6}$

$$y = \frac{28-16}{6} = \frac{12}{6} = 2$$

$$(x, y, z) = \left(\frac{10+t}{3}, \frac{28-8t}{6}, t\right)$$

$$(x, y, z) = (4, 2, 2)$$

Total = 4 Spider-man + 2 hulk + 2 Wonder woman

= 8 toys

Robert has to buy 8 toys out of which 4 has to be Spider-man, 2 Hulk, 2 Wonder woman to complete in a budget of \$60.

QUESTION#2

$A(1,1)$, $B(4,0)$, $C(2,3)$

i) Shearing along y-axis by factor 2
ii) Reflecting along $y=x$

i) Shearing along y-axis
$k=2$

$$A=\begin{bmatrix}1 & 0\\ k & 1\end{bmatrix} \rightarrow \begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}$$

Let $T(\vec{x}) = A\vec{x}$

): For vertice A : $A\vec{x} = \begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}\begin{bmatrix}1\\ 1\end{bmatrix} = \begin{bmatrix}1\\ 2+1\end{bmatrix} = \begin{bmatrix}1\\ 3\end{bmatrix}$

Let

= For vertice B: $A\vec{x} = \begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}\begin{bmatrix}4\\ 0\end{bmatrix} = \begin{bmatrix}4\\ 8\end{bmatrix}$

Let

= For vertice C : $A\vec{x} = \begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}\begin{bmatrix}2\\ 3\end{bmatrix} = \begin{bmatrix}2\\ 4+3\end{bmatrix} = \begin{bmatrix}2\\ 7\end{bmatrix}$

ii) Reflection along $y=x$

$$A=\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$$

$T(\vec{x}) = A\vec{x}$

= For D : $A\vec{x} = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}1\\ 3\end{bmatrix} = \begin{bmatrix}3\\ 1\end{bmatrix}$

= For E : $A\vec{x} = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}4\\ 8\end{bmatrix} = \begin{bmatrix}8\\ 4\end{bmatrix}$

$I =$ For $F = A\vec{x}' = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 7 \end{bmatrix} = \begin{bmatrix} 7 \\ 2 \end{bmatrix}$

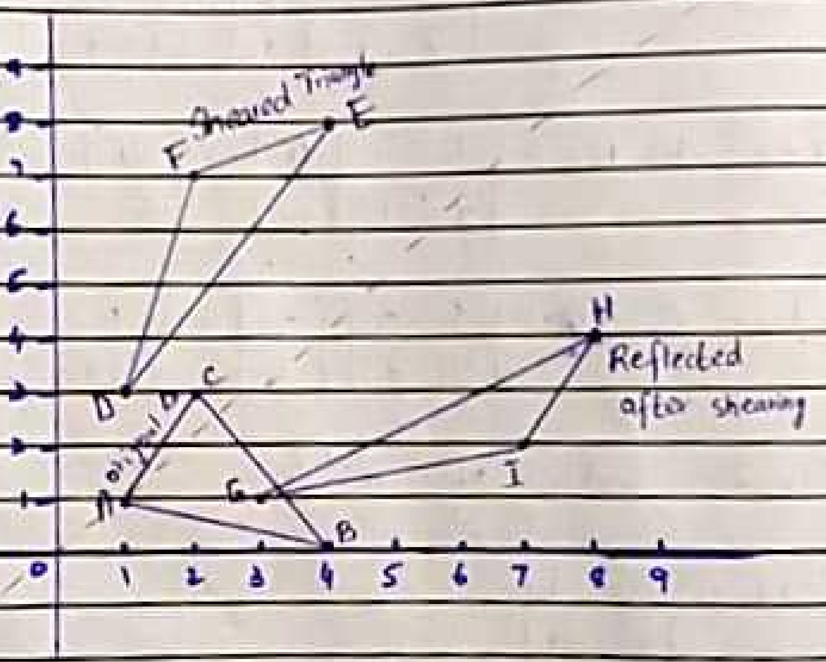
After shearing vertices are:

D(1, 3) , E(4, 8) , F(2, 7)

After Reflecting the sheared triangle vertices are:

G(3, 1) , H(8, 4) , I(7, 2)

Graph:-



Y = x

QUESTION # 3

(i)

a) $\hat{u}_1 + \hat{u}_2$

$\hat{u}_1 = (4, -1, 8)$ $\hat{u}_2 = (3, 2, -5)$

using standard property of addition

$\hat{u}_1 + \hat{u}_2 = (x_1, y_1, z_1) + (x_2, y_2, z_2)$
$= (x_1 + x_2 + 1, y_1 + y_2 - 1, z_1 + z_2)$

$= (4 + 3 + 1, -1 + 2 - 1, 8 - 5)$
$\hat{u}_1 + \hat{u}_2 = (8, 0, 3)$

b) $k\hat{u}_1$
$\hat{u}_1 = (4, -1, 8)$ and $k = 2$
using standard property of multiplication

$k\hat{u}_1 = k(x_1, y_1, z_1) = (0, ky_1, kz_1)$

$= 2(4, -1, 8) = (0, 2(-1), 2(8))$
$k\hat{u}_1 = (0, -2, 16)$

c) Zero vector
$\hat{u} + \hat{0} = (x_1, y_1, z_1) + (0, 0, 0)$

$= (x_1, y_1, z_1)$

$= \hat{u}$

d) Additive inverse :- vector $= u = (2, 3, 4)$

$\hat{u} + (-\hat{u}) = (2, 3, 4) + (-2, -3, -4)$

$= (2-2, 3-3, 4-4)$

$= (0, 0, 0)$

$= \hat{0}$

e) $k(\hat{u}_1 + \hat{u}_2) = k\hat{u}_1 + k\hat{u}_2$

LHS $= k(\hat{u}_1 + \hat{u}_2)$

$= k[(x_1, y_1, z_1) + (x_2, y_2, z_2)]$

$= k(x_1 + x_2 + 1, y_1 + y_2 - 1, z_1 + z_2)$

$= (kx_1 + kx_2 + k, ky_1 + ky_2 - k, kz_1 + kz_2)$

$= (0, ky_1 + ky_2 - k, kz_1 + kz_2)$

$= (0, ky_1, kz_1) + (0, ky_2 - k, kz_2)$

$= k(0, y_1, z_1) + k(0, y_2 - 1, z_2)$

$\neq k\hat{u}_1 + k\hat{u}_2$

Hence

LHS $\neq$ RHS

Property disproved.

f) $1\hat{u} = \hat{u}$

LHS $= (1, 1, 1)(x_1, y_1, z_1)$

$= (0, y_1, z_1)$ according to standard rule of multiplication.

LHS $\neq \hat{u}$

Hence $1 \cdot \hat{u} \neq \hat{u}$

LHS $\neq$ RHS

Property disproved.

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(ii)

a) $W_1 = \{(x, y),\ x \in R,\ y = -2x\}$

To find out the subset is subspace two properties are to be satisfied.

1) Closed under addition
2) Closed under multiplication

Let $\hat{u} = (u_1, u_1)$ $\quad \hat{v} = (v_1, v_1)$

as $y = -2x$ hence

$\hat{u} = (u_1, -2u_1)$ $\quad \hat{v} = (v_1, -2v_1)$

1) $\hat{u} + \hat{v} = (u_1, -2u_1) + (v_1, -2v_1)$
$= (u_1 + v_1, -2u_1 - 2v_1)$
$= (u_1 + v_1, -2(u_1 + v_1)) \in W_1$ (satisfied)

2) $k\hat{u} = k(u_1, -2u_1)$
$= (ku_1, -2ku_1) \in W_1$ (satisfied)

As both properties are satisfied hence W_1 subset is a vector subspace.

b) $W_2 = \{(x, y) : x = y^2,\ y \in R\}$

Let $\hat{u} = (u_2^2, u_2)$ $\quad \hat{v} = (v_2^2, v_1)$

1) $\hat{u} + \hat{v} = (u_2^2, u_2) + (v_2^2, v_1)$
$(u_2^2 + v_2^2, u_2 + v_2) \notin W_2$ (not satisfied)

2) $k\hat{u} = k(u_2^2, u_2)$
$= (ku_2^2, ku_2) \in W_2$ (satisfied)

As the two properties approve hence W_2 is not a subspace of V

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QUESTION # 4

(i)

$p = x^2 - 1$, $q = x^2 + x + 2$, $r = -4x^2 - 2x - 2$

$\Rightarrow k_1 v_1 + k_2 v_2 + k_3 v_3 = 0$

$k_1 p + k_2 q + k_3 r = 0$

$k_1(x^2-1) + k_2(x^2+x+2) + k_3(-4x^2-2x-2) = 0$

$(k_1 x^2 - k_1) + (k_2 x^2 + k_2 x + 2k_2) + (-4k_3 x^2 - 2k_3 x - 2k_3) = 0$

$(k_1 x^2 + k_2 x^2 - 4k_3 x^2) + (0 + k_2 x - 2k_3 x) + (-k_1 + 2k_2 - 2k_3) = 0$

$x^2(k_1 + k_2 - 4k_3) + x(k_2 - 2k_3) + (-k_1 + 2k_2 - 2k_3) = 0$

$$k_1 + k_2 - 4k_3 = 0$$
$$k_2 - 2k_3 = 0$$
$$-k_1 + 2k_2 - 2k_3 = 0$$

In augmented matrix form

$$\left[\begin{array}{ccc|c} 1 & 1 & -4 & 0 \\ 0 & 1 & -2 & 0 \\ -1 & 2 & -2 & 0 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -4 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 3 & -6 & 0 \end{array}\right] \quad R_3 + R_1$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \begin{array}{l} R_1 - R_2 \\ R_3 - 3R_2 \end{array}$$

$$\begin{array}{ccc} 0 & 3 & -6 \\ 0 & -3 & 6 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$k_1 - 2k_3 = 0 \Rightarrow k_1 = +2t$

$k_2 - 2k_3 = 0 \Rightarrow k_2 = 2t$

let $k_3 = t$

$k_1 = 2t$, $k_2 = 2t$ $k_3 = t$

As k_1, k_2, k_3 are all not equal to zero hence p, q, r are linearly dependent vectors.

(ii)

Span:–

$S = \{(1,2,3), (1,4,6), (2,-3,-5)\}$

To find span we will find the determinant

Let $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & -3 \\ 3 & 6 & -5 \end{bmatrix}$

$\det(A) = 1(-20+18) - 1(-10+9) + 2(12-12)$
$= 1(-2) - 1(-1) + 2(0)$
$= -2 + 1$
$= -1 \neq 0$

S spans R^3

Linear Combination

$v = (9, 1, 0)$

$$w = k_1 v_1 + k_2 v_2 + k_3 v_3$$

$$(9, 1, 0) = k_1(1, 2, 3) + k_2(1, 4, 6) + k_3(2, -3, -5)$$

$$= (k_1, 2k_1, 3k_1) + (k_2, 4k_2, 6k_2) + (2k_3, -3k_3, -5k_3)$$

$$(9, 1, 0) = (k_1 + k_2 + 2k_3), (2k_1 + 4k_2 - 3k_3), (3k_1 + 6k_2 - 5k_3) \quad -(1)$$

$$k_1 + k_2 + 2k_3 = 9$$
$$2k_1 + 4k_2 - 3k_3 = 1$$
$$3k_1 + 6k_2 - 5k_3 = 0$$

In augmented matrix form

$$\left[\begin{array}{ccc:c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array}\right]$$

$$\left[\begin{array}{ccc:c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{array}\right] \quad \begin{array}{l} R_2 - 2R_1 \\ R_3 - 3R_1 \end{array} \quad \boxed{\begin{array}{cccc} 2 & 4 & -3 & 1 \\ -2 & -2 & -4 & -18 \\ 3 & 6 & -5 & 0 \\ -3 & -3 & -6 & -27 \end{array}}$$

$$\left[\begin{array}{ccc:c} 1 & 1 & 2 & 9 \\ 0 & -1 & 4 & 10 \\ 0 & 3 & -11 & -27 \end{array}\right] \quad R_2 - R_3 \quad \boxed{\begin{array}{cccc} 0 & 2 & -7 & -17 \\ 0 & -3 & 11 & 27 \end{array}}$$

$$\left[\begin{array}{ccc:c} 1 & 1 & 2 & 9 \\ 0 & 1 & -4 & -10 \\ 0 & 3 & -11 & -27 \end{array}\right] \quad R_2 \times \frac{-1}{1}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 6 & 19 \\ 0 & 1 & -4 & -10 \\ 0 & 0 & 1 & 3 \end{array}\right] \quad \begin{array}{l} R_1 - R_2 \\ R_3 - 3R_2 \end{array}$$

$$\begin{array}{cccc} 1 & 1 & 2 & 9 \\ 0 & -1 & 4 & 10 \\ \hline 0 & 3 & -11 & -27 \\ 0 & -3 & 12 & 30 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right] \quad \begin{array}{l} R_2 + 4R_3 \\ R_1 - 6R_3 \end{array}$$

$$\begin{array}{cccc} 0 & 1 & -4 & -10 \\ 0 & 0 & 4 & 12 \\ \hline 1 & 0 & 6 & 19 \\ 0 & 0 & -6 & -18 \end{array}$$

$k_1 = 1, \ k_2 = 2, \ k_3 = 3$

putting in eq ①

$$(9, 1, 0) = (1 + 2 + 2(3)), \ (2(1) + 4(2) - 3(3)), \ (3(1) + 6(2) - 5(3))$$

$$= (3+6), \ (2+8-9), \ (3+12-15)$$

$$(9, 1, 0) = (9, 1, 0)$$

Hence it is a Linear Combination

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QUESTION #5

$$A = \begin{bmatrix} -2 & 0 & 2 \\ -2 & 1 & 2 \\ 4 & 0 & 5 \end{bmatrix}$$

To find eigenvalues

$$\det(A - \lambda I) = 0$$

$$A - \lambda I = \begin{bmatrix} -2 & 0 & 2 \\ -2 & 1 & 2 \\ 4 & 0 & 5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 0 & 2 \\ -2 & 1 & 2 \\ 4 & 0 & 5 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} -2-\lambda & 0 & 2 \\ -2 & 1-\lambda & 2 \\ 4 & 0 & 5-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} -2-\lambda & 0 & 2 \\ -2 & 1-\lambda & 2 \\ 4 & 0 & 5-\lambda \end{vmatrix}$$

$$= -2-\lambda\big((1-\lambda)(5-\lambda) - 0\big) - 0\big((-2)(5-\lambda) - 8\big) + 2\big(0 - 4(1-\lambda)\big)$$

$$= -2-\lambda\,(5 - \lambda - 5\lambda + \lambda^2) - 0 + 2\,(-4 + 4\lambda)$$

$$= -10 + 2\lambda + 10\lambda - 2\lambda^2 - 5\lambda + \lambda^2 + 5\lambda^2 - \lambda^3 - 8 + 8\lambda$$

$$= -\lambda^3 + 4\lambda^2 + 15\lambda - 18$$

$$= \lambda^3 - 4\lambda^2 - 15\lambda + 18$$

	(3)	(2)	(1)	(0)
	1	-4	-15	18
1	↓	1	-3	-18
	1	-3	-18	$\boxed{0}$

$(\lambda-1)(\lambda^2-3\lambda-18)=0$

$\lambda-1=0$ $\qquad$ $\lambda^2-3\lambda-18=0$

$\lambda=1$ $\qquad$ $\lambda^2-6\lambda+3\lambda-18=0$

$\lambda(\lambda-6)+3(\lambda-6)=0$

$(\lambda-6)(\lambda+3)=0$

$\lambda-6=0 \qquad \lambda+3=0$

$\lambda=6 \qquad \lambda=3$

$\lambda=1,6,-3$

To find eigen vecto

$(A-\lambda I)\vec{x}=0$

for $\lambda=-3$

$$\begin{bmatrix} -2-\lambda & 0 & 2 \\ -2 & 1-\lambda & 2 \\ 4 & 0 & 5-\lambda \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -2-(-3) & 0 & 2 \\ -2 & 1-(-3) & 2 \\ 4 & 0 & 5-(-3) \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 \\ -2 & 4 & 2 \\ 4 & 0 & 8 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x + 2z = 0$

$-2x + 4y + 2z = 0$

$4x + 8z = 0$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ -2 & 4 & 2 & 0 \\ 4 & 0 & 8 & 0 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 4 & 6 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} \\ R_2 + 2R_1 \\ R_3 - 4R_1 \end{array}$$

$$\begin{array}{ccc} -2 & 4 & 2 \\ 2 & 0 & 4 \\ \hline 4 & 0 & 8 \\ -4 & 0 & -8 \end{array}$$

$x + 2z = 0$
$4y + 6z = 0$

Let $z = t$

$x = -2z \Rightarrow -2t$
$4y = -6z \Rightarrow -6t$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2t \\ \frac{-6t}{4} \\ t \end{bmatrix} \Rightarrow t \begin{bmatrix} -2 \\ -6/4 \\ 1 \end{bmatrix}$$

Eigenvector for negative value i.e $\lambda = -3$ is

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \begin{bmatrix} -2 \\ -6/4 \\ 1 \end{bmatrix} \quad -3/2$$